

```
Terminal x
Welcome to Ubuntu 22.04.2 LTS (GNU/Linux 5.19.0-1028-aws x86_64)

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 *
 * Welcome to the Codio Terminal!
 *
 * https://docs.codio.com/develop/develop/ide/boxes/overview
 *
 * Your Codio Box domain is: tapeford-dependadrian.codio.io
 *
Last login: Mon May 22 11:11:24 2023 from 192.168.10.93
codio@tapeford-dependadrian:~/workspace$ mkdir my_project
codio@tapeford-dependadrian:~/workspace$ cd my_project
codio@tapeford-dependadrian:~/workspace/my_project$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:     git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:     git branch -m <name>
Initialized empty Git repository in /home/codio/workspace/my_project/.git/
codio@tapeford-dependadrian:~/workspace/my_project$ mkdir -p .github/workflows
codio@tapeford-dependadrian:~/workspace/my_project$ touch .github/workflows/chatgpt_integration.yml
codio@tapeford-dependadrian:~/workspace/my_project$ touch chatgpt_integration.py
codio@tapeford-dependadrian:~/workspace/my_project$ git add .
codio@tapeford-dependadrian:~/workspace/my_project$
codio@tapeford-dependadrian:~/workspace/my_project$
codio@tapeford-dependadrian:~/workspace/my_project$ git commit -m "Initial commit with ChatGPT integration."
[master (root-commit) 645bcb0] Initial commit with ChatGPT integration.
```

ChatGPT API with project management and collaboration tools

We will explore how to integrate the ChatGPT API with **GitHub** to improve collaboration and streamline workflows. GitHub, a popular platform for version control and project management, can benefit from ChatGPT's capabilities to enhance various aspects of project management and development. This integration can provide intelligent suggestions for code, help with documentation, and assist in code review processes. In this tutorial, we will demonstrate how to use the ChatGPT API within a GitHub Actions workflow to automate some of these tasks.

Step 1: Create a New Repository

Create a new directory on your local machine and initialize it as a Git repository:

Use the terminal for the following:

```
mkdir my_project
cd my_project
git init
```

Step 2: Create the GitHub Actions Workflow

1. In your local Git repository, create a directory for GitHub Actions:

```
mkdir -p .github/workflows
```

2. Create a new YAML file for the ChatGPT integration workflow:

```
touch .github/workflows/chatgpt_integration.yml
```

For this page as you run the `touch` command to run your code, it will create a new file. You can access those file by clicking on the file button ->open file then you type in the file name you want to open. In the following case it should be `chatgpt_integration.yml`.

Once Open paste the following inside the text area:

```
name: ChatGPT Integration

on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main

jobs:
  chatgpt_integration:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout code
        uses: actions/checkout@v2

      - name: Set up Python
        uses: actions/setup-python@v2
        with:
          python-version: 3.9

      - name: Install dependencies
        run: |
          python -m pip install --upgrade pip
          pip install openai

      - name: Run ChatGPT API integration
        run: python chatgpt_integration.py
        env:
          OPENAI_KEY: "${{ secrets.OPENAI_KEY }}"
```

Step 3: Create the ChatGPT API Integration Script

1. In your local Git repository, create a new Python file:

```
touch chatgpt_integration.py
```

1. Open the `chatgpt_integration.py` file and add the following content:

```
import openai
import os

# Initialize the OpenAI API
openai.api_key = os.getenv('OPENAI_KEY')

def chatgpt_prompt(prompt):
    response = openai.ChatCompletion.create(
        model="gpt-3.5-turbo",
        messages=[
            {"role": "system", "content": "You are a helpful assistant."},
            {"role": "user", "content": prompt},
        ],
    )
    return (response['choices'][0]['message']['content']).strip()
#assistant_message = [msg for msg in response['choices'][0]['messages'] if msg['role'] == 'assistant']
#return assistant_message[0]['content']

# Example: Get a code suggestion from ChatGPT
code_suggestion_prompt = "Write a Python function to calculate the factorial of a number."
code_suggestion = chatgpt_prompt(code_suggestion_prompt)
print("Code Suggestion:")
print(code_suggestion)
```

Step 4: Commit Changes and Push to GitHub

1. Stage the files for commit:

```
git add .
```

- 2.Commit changes

```
git commit -m "Initial commit with ChatGPT integration."
```

3. Create a new repository on GitHub. You can do this by visiting <https://github.com/new> and following the instructions. Do not initialize the new repository with a README, .gitignore, or License. This empty repository will receive the contents of your local repository.
4. Once your new GitHub repository is created, you'll be shown a repository URL. It will look like this: <https://github.com/<username>/<repository>>. Copy this URL, replace and with your GitHub username and repository name, and run the following commands:

```
git remote add origin git@github.com:<username>:<repository>.git
git branch -M main
git push -u origin main
```

You now have a new GitHub repository with a GitHub Actions workflow that integrates with the ChatGPT API! Please remember to add your OpenAI API key to the GitHub repository secrets. You can do this by going to your GitHub repository, clicking on **"Settings"** -> **"Secrets"** -> **"New repository secret"**, then adding **"OPENAI_KEY"** as the name and your actual OpenAI API key as the value.

Please note that to use SSH with GitHub, you must have an SSH key set up on your GitHub account. You can find out how to do this in the GitHub Docs [here](#).

In the integration process, the OpenAI API key is stored in a Python file in the local repository.

☐ True

☒ False

The OpenAI API key is not stored in the local Python file. Instead, it's fetched from the environment variables which are set in the GitHub repository secrets.

Check it!

Next